

RESEARCH DESIGN, DATASETS & METHODOLOGICAL FRAMEWORK

Integrated Bioeconomic Evaluation: Census Panel, Non-Parametric Frontiers, and Spatial-Temporal Econometrics

PILLAR 1: DATASETS & EMPIRICAL PANEL

TURKSTAT & SUBIS Census Panel (2000–2024, N = 125)

Panel Structure & Regional Coverage

- Five maritime basins: Black Sea, Marmara, Aegean, Mediterranean, and Inland
- 25 consecutive annual observation periods (balanced panel of 125 units)
- Consistent spatial administrative boundaries across all census survey waves

Physical Capital & Fleet Dimensions

- Active commercial vessel counts disaggregated across vessel length classes
- Cumulative engine power in installed kilowatts and engine horsepower
- Cumulative gross tonnage reflecting physical vessel volume and hold capacity

Capture Production & Economic Value

- Annual marine capture harvest disaggregated into pelagic and demersal fish
- Landed catch revenue deflated to constant 2015 national currency levels
- Controls for temporal price shifts, species mix variation, and inflation

Technology Adoption & Policy Interventions

- Equipment adoption rates: GPS, echo sounders, radar, sonar, and VHF radio
- 2002 National Commercial Fishing License Freeze (entry moratorium)
- 2012–2018 State Vessel Buyback Scheme (targeted fleet decommissioning)

PILLAR 2: FRONTIER MODELING & CAPACITY

Unsupervised PCA, DEA Frontiers & Malmquist TFP

Modernization Index Construction

- Unsupervised Principal Component Analysis on five gear technologies
- First principal component extraction capturing dominant common variance
- Min-max standardization providing orthogonal continuous technology index

Data Envelopment Analysis (DEA)

- Non-parametric output-oriented linear programming capacity frontiers
- Dual benchmarking under Constant and Variable Returns to Scale assumptions
- Capacity utilization defined as observed output relative to frontier potential

Scale Efficiency Decomposition

- Scale efficiency ratio derived from dual frontier benchmark specifications
- Operational returns classification into optimal, increasing, or decreasing scale
- Separates capacity gap into operational scale distortion versus inefficiency

Productivity & Finite-Sample Correction

- Dynamic Malmquist TFP index: technical efficiency change and frontier shifts
- Homogeneous smoothed bootstrap resamples for finite-sample bias correction
- Constructs robust confidence intervals for efficiency and productivity metrics

PILLAR 3: ECONOMETRICS & IDENTIFICATION

Spatial Diagnostics, Robust HAC & Sensitivity Suite

Pre-Estimation Diagnostic Protocol

- Cross-sectional dependence tests via Pesaran CD and Friedman procedures
- Panel serial autocorrelation checks using Wooldridge and Durbin–Watson tests
- Panel unit root and stationarity validation via Im–Pesaran–Shin tests

Spatial & Serial HAC Estimation

- Driscoll–Kraay non-parametric HAC standard error covariance estimator
- Inference robust to arbitrary spatial dependence and serial correlation
- Prais–Winsten panel generalized least squares with serial quasi-differencing

Two-Way Fixed Effects Specifications

- Panel specifications with maritime basin and annual observation fixed effects
- Controls for time-invariant regional unobservables and aggregate macro shocks
- Estimates modernization elasticity, capital responsiveness, and policy shifts

Alternative Estimators & Sensitivity Suite

- Censored Tobit and Fractional Logit QMLE for bounded capacity outcomes
- Leave-one-basin-out Jackknife iterations verifying spatial parameter stability
- Non-linear Machine Learning Random Forest and SHAP feature importance

SEQUENTIAL
PIPELINE

Step 1: Census Ingestion
TURKSTAT & SUBIS (N = 125)

Step 2: Technology Indexing
PCA Modernization Index

Step 3: Frontier Modeling
DEA Frontiers & Malmquist TFP

Step 4: Econometric ID
Driscoll–Kraay & AR(1) FGLS

Step 5: Robustness Suite
Tobit, FracLogit, Jackknife & ML